



Chapter 2.1

Introduction to Fuzzy Sets



Uncertainty

- According to the traditional view of science, science should strive for certainty in all its manifestations (precision, specificity, sharpness, consistency, etc.)
- Uncertainty (imprecision, non-specificity, vagueness, inconsistency, etc.) is regarded as unscientific.
- According to the alternative (or modern) view, uncertainty is considered essential to science.



Model

- In constructing a model, we always attempt to maximize its usefulness.
- This aim is closely connected with the relationship among three key characteristics of every systems model: **complexity**, **credibility**, and **uncertainty**
- In general, allowing more uncertainty tends to reduce complexity and increase credibility of the resulting model.

There is a **trade-off**:

If you try to **remove uncertainty completely**:

- You add more parameters
- Model becomes **complex**
- May reduce interpretability → lower credibility

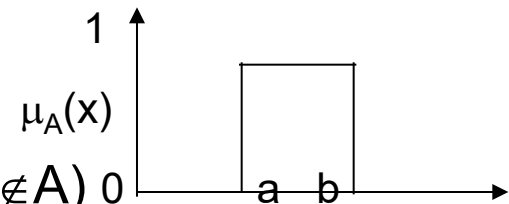
If you **allow uncertainty** (e.g., probabilistic or fuzzy models):

- Model becomes **simpler**
- More **robust and realistic**
- Often **more credible**

Crisp/ Classical/ Conventional Sets



- A **Universal set** (represented by X) is a set consisting of all possible elements (also known as members) related to a particular context.
- A **set** is generally represented by its elements.
- $A = \{a_1, a_2, \dots, a_n\}$
- A set can also be represented by the **property** of its element like the following
 $A = \{x | P(x)\},$
- That is, set. A contains all x belonging to X that have the property P .
- A set A can also be indicated by its **characteristic function** as follows:
 - $\mu_A(x) = 1$ (true), if x belongs to $A, (x \in A)$
 - $\mu_A(x) = 0$ (false), if x does not belong to $A, (x \notin A)$



Notations Used in Set Theory



1. ϕ : It indicates an **empty set** (also known as **null set**) that does not contain any element.
2. $x \in A$: It states that an element x of the Universal set X **belongs to** set A .
3. $x \notin A$: It represents a condition that an element x of the Universal set X **does not belong to** set A .
4. $A \subseteq B$: It is used to indicate that set **A is a subset of set B** , that is, each element of A is also an element of B but the reverse may not be true.
5. $A \supseteq B$: It indicates that set **A is a superset of set B** , that is, each element of B is also an element of A but the reverse may not be true.
6. $A = B$: It states that the sets **A and B are equal**, that is, both of them contain the same elements. If $A \subseteq B$ and $B \subseteq A$, then $A = B$.

Notations Used in Set Theory (Cont.)



7. $A \neq B$: It says that the sets A and B are not equal.
8. $A \subset B$: It is used to represent that A is a proper subset of B , for which both the conditions $A \subseteq B$ and $A \neq B$ are to be true and consequently, B contains at least one element that is missing in A .
9. $A \supset B$: It states that the set A is a proper superset of set B , for which both the conditions $A \supseteq B$ and $A \neq B$ are to be satisfied. It means that set A contains at least one element that is missing in set B .
10. $|A|$: It indicates the cardinality of set A that is defined as the total number of elements present in that set. For example, in case of a singleton set $A = \{a_1\}$, $|A| = 1$.
11. $p(A)$: It represents the power set of A , which is nothing but the maximum number of subsets including the null that can be constructed from a set A . Let us consider that set A contains three elements, that is, $A = \{a_1, a_2, a_3\}$. The subsets (8 in number), those can be constructed from the set A are: $\{a_1\}$, $\{a_2\}$, $\{a_3\}$, $\{a_1, a_2\}$, $\{a_1, a_3\}$, $\{a_2, a_3\}$, $\{a_1, a_2, a_3\}$, $\{\phi\}$. Now, the cardinality of set A , that is, $|A| = 3$ and that of $p(A)$ is given by $|p(A)| = 2^{|A|} = 2^3 = 8$.



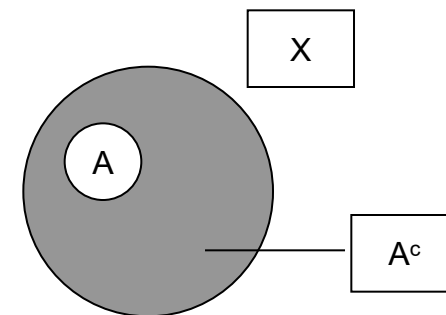
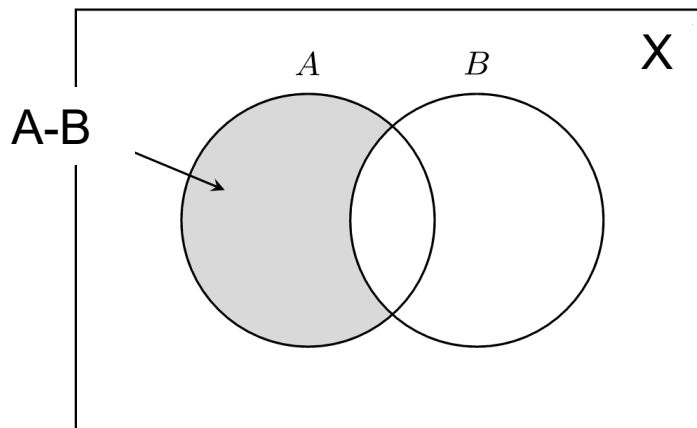
Crisp Set Operations

Difference: The difference between two sets A and B is represented as follows:

$$A-B = \{x | x \in A \text{ and } x \notin B\}$$

It is also known as **relative complement** of set B with respect to set A, which contains all the elements of A, those are not included in B. **Absolute complement** of a set A is indicated by A^c and it is defined as the difference between the Universal set X and set A as given below

$$A^c = X - A = \{x | x \in X \text{ and } x \notin A\}$$





Crisp Set Operations(Cont.)

Notes:

1. Complement of a complement yields the original set.
2. Absolute complement of an empty set is nothing but the Universal set.
3. Absolute complement of the Universal set yields an empty set.

Example: Let $A = \{a, c, d, f, r, g, h\}$ and $B = \{c, d, r, m, n\}$. Therefore, $A - B = \{a, f, g, h\}$.



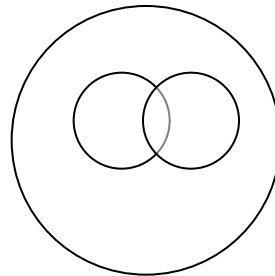
Crisp Set Operations (Cont.)

Intersection: Intersection of two sets, namely A and B is denoted by $A \cap B$ and it is defined as follows:

$$A \cap B = \{x | x \in A \text{ and } x \in B\}$$

Two sets A and B are called **disjoint**, if there is no common element between them and it is indicated by $A \cap B = \phi$

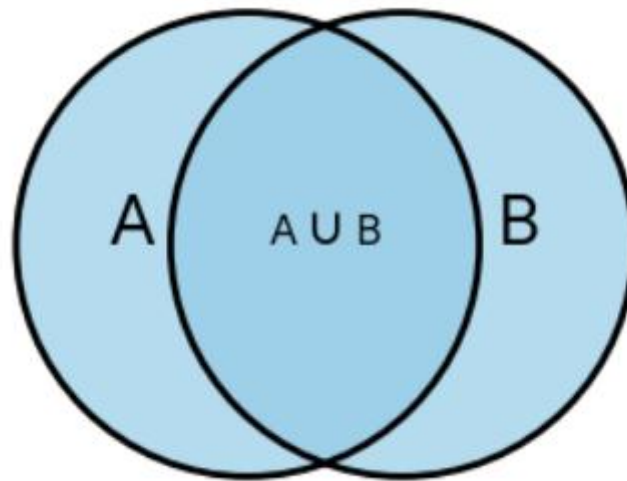
Example: Let $A = \{a, c, d, f, r, g, h\}$ and $B = \{c, d, r, m, n\}$. Therefore, $A \cap B = \{c, d, r\}$.





Crisp Set Operations (Cont.)

- **Union:** Union of two sets, such as A and B is represented by $A \cup B$ and it is defined like the following:
 - $A \cup B = \{x | x \in A \text{ or } x \in B\}$
 - The union of two sets A and B is shown in Figure as the shaded area.
 - Example: Let $A = \{a, c, d, f, r, g, h\}$ and $B = \{c, d, r, m, n\}$. Therefore, $A \cup B = \{a, c, d, f, r, g, h, c, d, r, m, n\}$.





Properties of Crisp Sets

Let us consider some crisp sets A, B and C defined in the Universal set X

1. Law of involution: $\overline{\overline{A}}=A$

2. Laws of commutativity: $A \cup B = B \cup A$

3. Laws of associativity: $(A \cup B) \cup C = A \cup (B \cup C)$

$$(A \cap B) \cap C = A \cap (B \cap C)$$

4. Laws of distributivity : $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

5. Laws of idempotence/tautology: $A \cup A = A$

$$A \cap A = A$$

6. Laws of absorption: $A \cup (A \cap B) = A$

$$A \cap (A \cup B) = A$$



Properties of Crisp Sets (Cont.)

7. Laws of identity: $A \cup X = X$

$$A \cap X = A$$

$$A \cup \phi = A$$

$$A \cap \phi = \phi$$

8. De Morgan's Laws: $\overline{(A \cup B)} = \bar{A} \cap \bar{B}$

$$\overline{(A \cap B)} = \bar{A} \cup \bar{B}$$

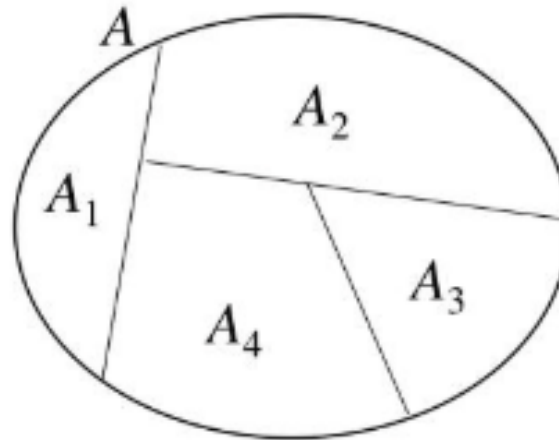
9. Law of contradiction: $A \cap \bar{A} = \phi$

10. Law of excluded middle: $A \cup \bar{A} = X$



Partition

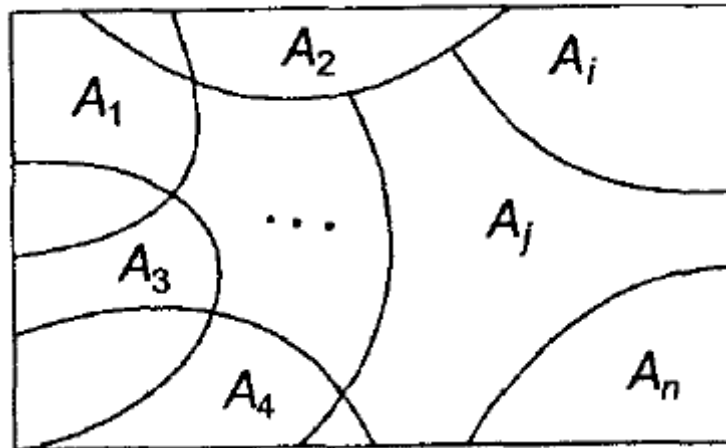
- A partition on A is defined to be a set of non-empty subsets A_i , each of which is **pairwise disjoint** and whose **union yields the original set A** .
- Partition on A indicated as $\pi(A)$, is therefore
 1. $A_i \cap A_j = \emptyset$ for each pair $(i,j) \in I, i \neq j$
 2. $\bigcup_{i \in I} A_i = A$





Covering

A-covering on A is defined to be a set of non-empty subsets A_i whose union yields the set A . The non-empty subsets need not be disjoint.





Rule of Addition

- Given a partition on A where A_i , $i = 1, 2, \dots, n$ are its non-empty subsets then,
- $|A| = |\bigcup_{i=1}^n A_i| = \sum_{i=1}^n |A_i|$



Rule of Inclusion and Exclusion

- Rule of addition is not applicable on the covering of set A , especially if the subsets are not pairwise disjoint. In such a case, the rule of inclusion and exclusion is applied
- Example: Given A to be a covering of n sets $A_1, A_2, \dots, A_n$

$$\text{For } n=2, |A|=|A_1 \cup A_2|=|A_1|+|A_2|-|A_1 \cap A_2|$$

$$\text{For } n=3, |A|=|A_1 \cup A_2 \cup A_3|=|A_1|+|A_2|+|A_3|-|A_1 \cap A_2|-|A_2 \cap A_3|-|A_3 \cap A_1|+|A_1 \cap A_2 \cap A_3|$$



Fuzzy Sets

- Fuzzy Sets are the sets with imprecise (vague) boundaries.
- Fuzziness occurs in everyday communication through natural language.
- Natural language contains vague concepts such as beautiful, tall, long, etc. which are understood by humans but cannot be modeled by ordinary set theory.
- Fuzziness is subjective as it is related to semantic of the word used.
- Subjectivity belongs more to psychology than science!
- Science means the study of laws of nature, not including laws of human mind?
- Such events were reduced to conventional mathematics with the expense of imprecision.
- How much information is lost this way, were not raised prior to 1965.



Fuzzy Sets (Cont.)

- In 1965, Zadeh [1] attacked the problem of modeling fuzzy concepts and a logic of fuzzy concepts from a mathematical viewpoint in his seminal paper.
- Although a related idea was visualized by Max Black, an American Philosopher, in 1937 [2].
- Prior to 1965, people used to consider **probability theory** (which works based on Aristotelian two-valued logic) as the prime agent for dealing with uncertainties.
- The above logic uses the concept of classical or crisp set theory.



Fuzzy Sets (Cont.)

- Fuzzy versus Probability
 - Both $\in [0,1]$
 - Fuzzy $\neq$ Probability
 - Probability theory based on frequency
 - Fuzzy set theory based on similarity
- Example:
 - X = Set of all liquids
 - A = Drinkable liquids
 - From which bottle should we drink?



$$\mu_A(P)=0.9 \quad \Pr(Q \in A)=0.9$$



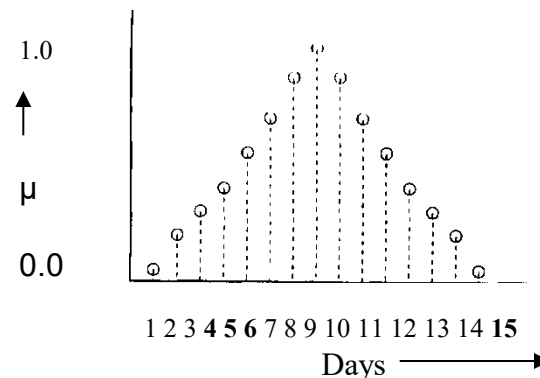
Fuzzy Sets (Cont.)

- Fuzzy Sets, $A(x)$ is represented by a pair of two things –
 - The element x and
 - Its membership value $\mu_A(x)$
- $A(x) = \{(x, \mu_A(x)), x \in X\}$
- Generalization of conventional sets
- It tries to model human thinking process
- Fuzzy sets contain objects that satisfy imprecise properties with varying degrees Fuzzy
- Membership function : $\mu_A(x) \in [0,1]$ signifies the grade of membership of x in A i.e., degree of possessing in some imprecise property represented by A function.
- It may be either discrete or continuous.



Discrete Fuzzy Sets

- It can be expressed using the membership values of its elements.
- $A(x) = \sum_{i=1}^n \mu_A(x_i) / x_i$
- Let us take an example, in which atmospheric temperature during the period of to 15-th day of a month has been found to be medium. The medium temperature can be represented:
- $M = 0.05/1 + 0.2/2 + 0.3/3 + 0.4/4 + 0.55/5 + 0.7/6 + 0.875/7 + 1.0/8 + 0.875/9 + 0.7/10 + 0.55/11 + 0.4/12 + 0.3/13 + 0.2/14 + 0.05/15.$





Continuous Fuzzy Sets

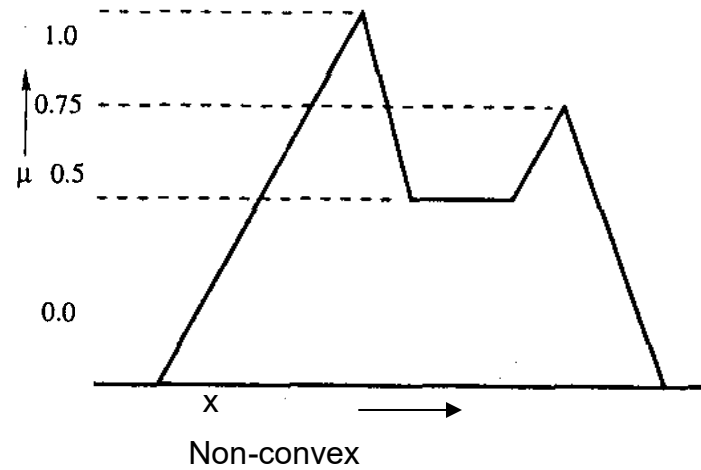
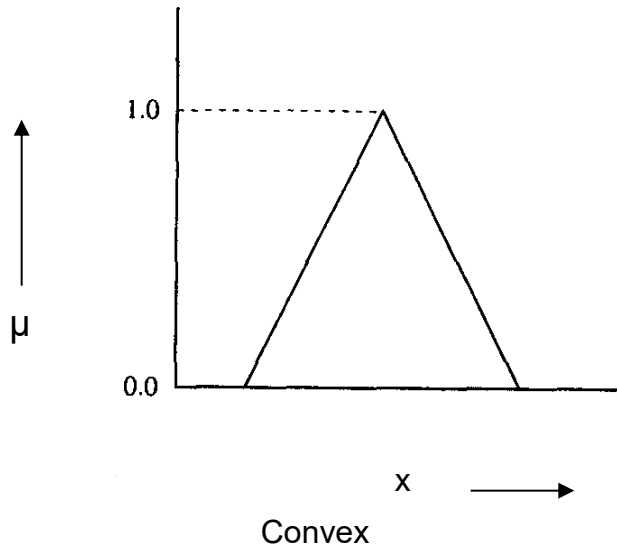
- $A(x) = \int_X \mu_A(x)/x$
- Integral sign is utilized here to indicate the collection of the pairs of elements and their membership values used to represent a continuous fuzzy set.
- It is represented with the help of a membership function distribution.
- **Convex fuzzy set:** A fuzzy set $A(x)$, such that $x \in X$ (where X is the Universal set) will be convex in nature, if the following condition is satisfied:

$$\mu_A\{\lambda x_1 + (1 - \lambda) x_2\} \geq \min\{\mu_A(x_1), \mu_A(x_2)\}$$

Where x_1 and x_2 are two values of x lying within X , λ is a constant lying in the range of (0.0, 1.0)



Continuous Fuzzy Sets (Cont.)





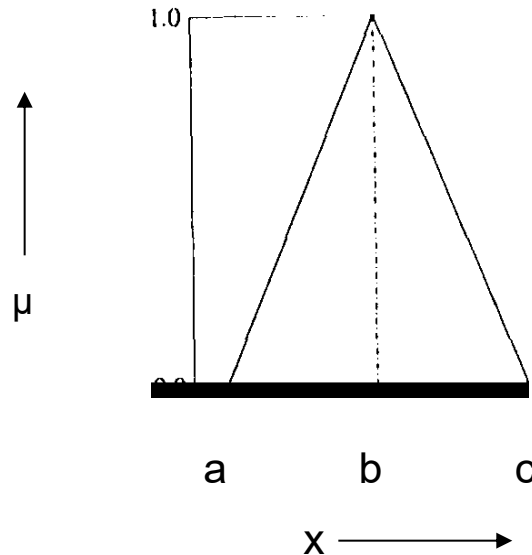
Fuzzy Membership Functions

- Transition of membership to non membership is **gradual** rather than abrupt.
- Assignment of membership functions is **subjective** in nature.
- Membership function of a fuzzy set may not be unique
- Source of membership functions: **data and people**



Triangular Membership Functions

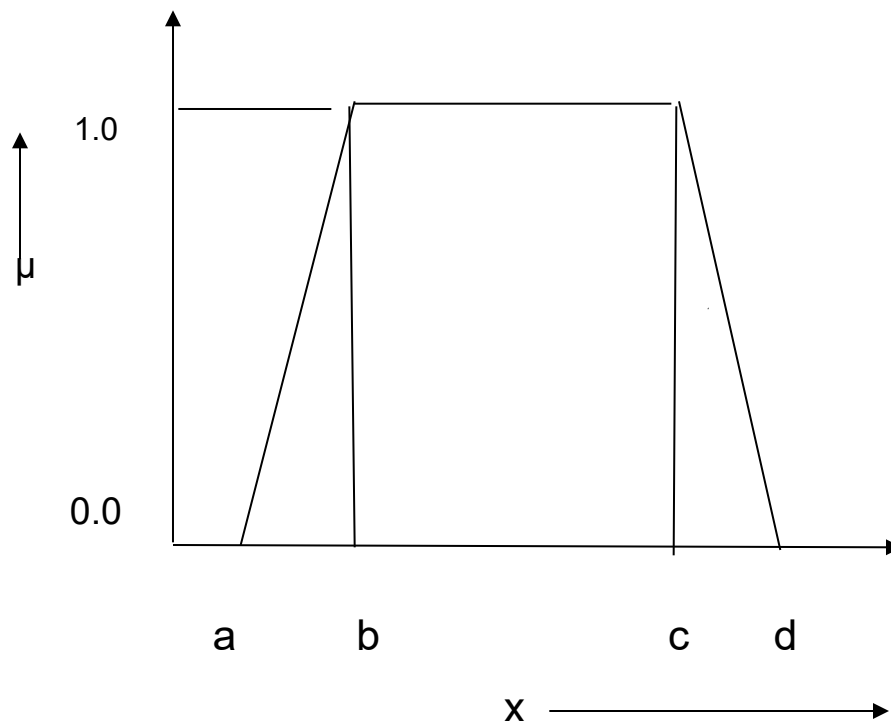
- **Triangular Membership** Function: Denoted by Triangle (x;a,b,c)
- $\mu_{\text{triangle}} = \max(\min((x-a)/(b-a), (c-x)/(c-b)), 0)$





Trapezoidal Membership Functions

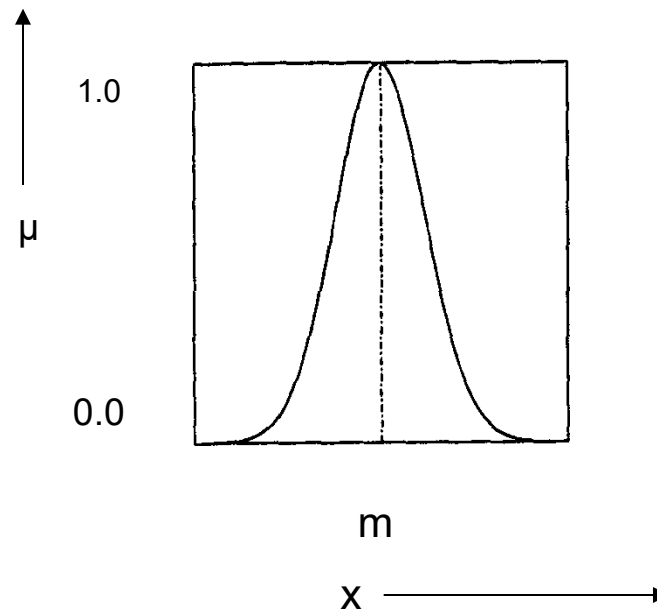
- Trapezoidal Membership Functions: Denoted by Trapezoidal $(x;a,b,c,d)$
- $\mu_{\text{Trapezoidal}} = \max(\min((x-a)/(b-a), 1, (d-x)/(d-c)), 0)$





Gaussian Membership Functions

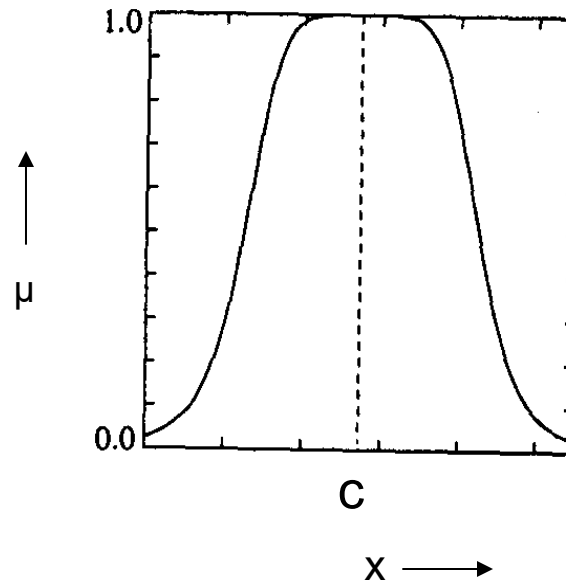
- Gaussian Membership Functions : Indicated by Gaussian (x ; m , σ), where m and σ represent the mean and standard deviation of the distribution, respectively.
- $\mu_{\text{Gaussian}} = \frac{1}{\sigma \sqrt{2\pi}} e^{-0.5((x-m)/\sigma)^2}$



Bell-shaped Membership Function



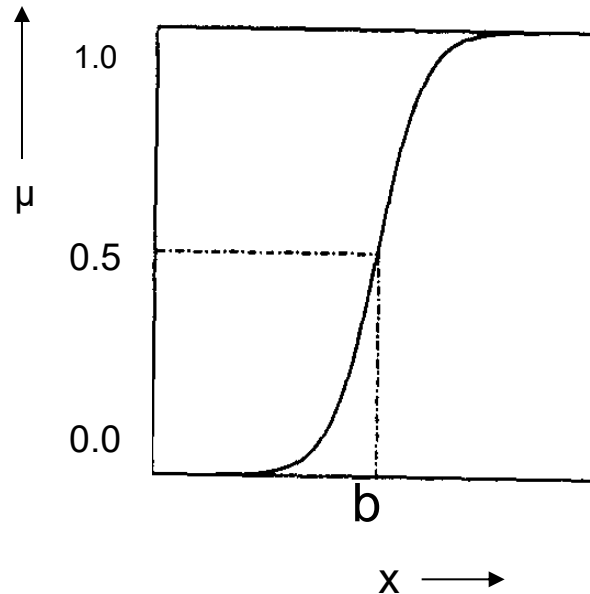
- Bell-shaped Membership Function : Represented as Bell-shaped ($x; a, b, c$)
- $\mu_{\text{Bell-shaped}} = 1/(1+|x-c/a|^{2b})$





Sigmoid Membership Function

- Sigmoid Membership Function : Denoted by Sigmoid (x; a, b)
- $\mu_{\text{Sigmoid}} = 1/(1+e^{-a(x-b)})$



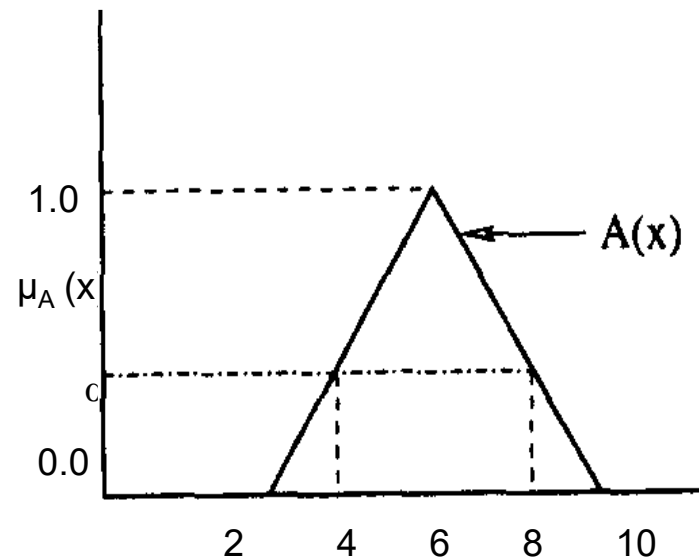


A Few Definitions in Fuzzy Sets



α -cut

- α -cut of a fuzzy set $A(x)$: It is a set consisting of elements x of the Universal set X , whose membership values are either greater than or equal to the value of α
- $\alpha_{\mu A(x)} = \{x | \mu_A(x) \geq \alpha\}$
- Strong α -cut of a fuzzy set $A(x)$:
- $\alpha^+_{\mu A(x)} = \{x | \mu_A(x) > \alpha\}$





Support

- Support of a fuzzy set $A(x)$: set of all $x \in X$, such that $\mu_A(x) > 0$.
- $\text{supp}(A) = \{x \in X \mid \mu_A(x) > 0\}$
- It is nothing but strong 0-cut.



Scalar cardinality

- Scalar cardinality of a fuzzy set $A(x)$:
- $|A(x)| = \sum_{x \in X} \mu_A(x)$
- Example:
 - Let $A(x) = (X_1, 0.1), (X_2, 0.2), (X_3, 0.3), (X_4, 0.4)$
 - Its cardinality $|A(x)| = 0.1 + 0.2 + 0.3 + 0.4 = 1.0$.



Some more definitions

- **Core** of a fuzzy set $A(x)$: It is nothing but its 1-cut.
- **Height** of a fuzzy set $A(x)$: It is denoted by $h(A)$ and defined as the largest of membership values of the elements contained in that set.
- **Normal fuzzy set**: A fuzzy set $A(x)$ is called normal, if its height $h(A) = 1.0$.
- **Sub-normal fuzzy set**: A fuzzy set $A(x)$ is known as a sub-normal fuzzy set, if height $h(A) < 1.0$



Some Standard Operations in Fuzzy Sets



Subset of a fuzzy set

- The fuzzy set $A(x)$ is called the subset of $B(x)$, if $\mu_A(x) < \mu_B(x)$
- $A(x) \subset B(x)$, if $\mu_A(x) < \mu_B(x)$

- **Example:**

Let us consider the following two fuzzy sets:

$$A(x) = \{(X_1, 0.1), (X_2, 0.2), (X_3, 0.3), (X_4, 0.4)\}$$

$$B(x) = \{(X_1, 0.5), (X_2, 0.7), (X_3, 0.8), (X_4, 0.9)\}$$

As $\mu_A(x) < \mu_B(x)$ for all $x \in X$, $A(x) \subset B(x)$



Equal fuzzy sets

- $A(x)=B(x)$, if $\mu_A(x) = \mu_B(x)$

- **Example:**

Let us consider the following two fuzzy sets:

$$A(x) = \{(X_1, 0.1), (X_2, 0.2), (X_3, 0.3), (X_4, 0.4)\}$$

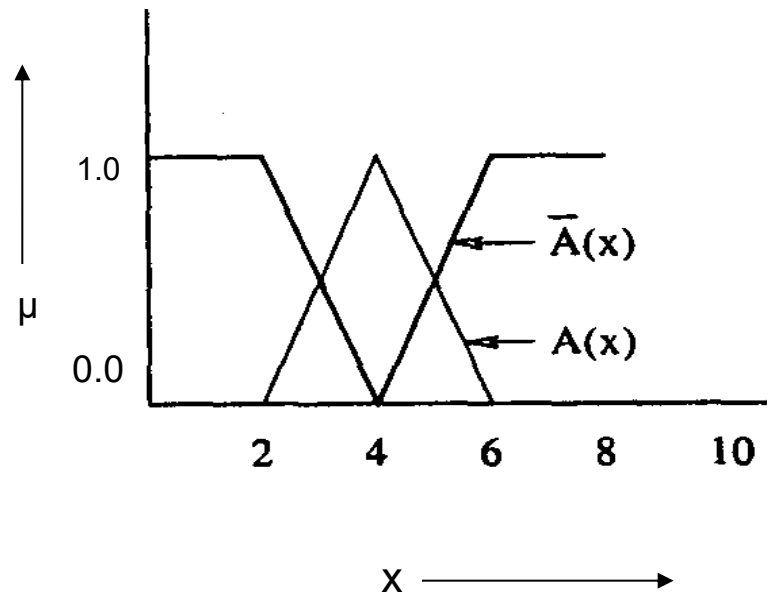
$$B(x) = \{(X_1, 0.5), (X_2, 0.7), (X_3, 0.8), (X_4, 0.9)\}$$

As $\mu_A(x) \neq \mu_B(x)$ for different $x \in X$, $A(x) \neq B(x)$



Complement of a fuzzy set

- $\bar{A}(x) = 1 - A(x)$, for all $x \in X$.





Intersection of fuzzy sets

- $\mu_{(A \cap B)}(x) = \min\{\mu_A(x), \mu_B(x)\}$

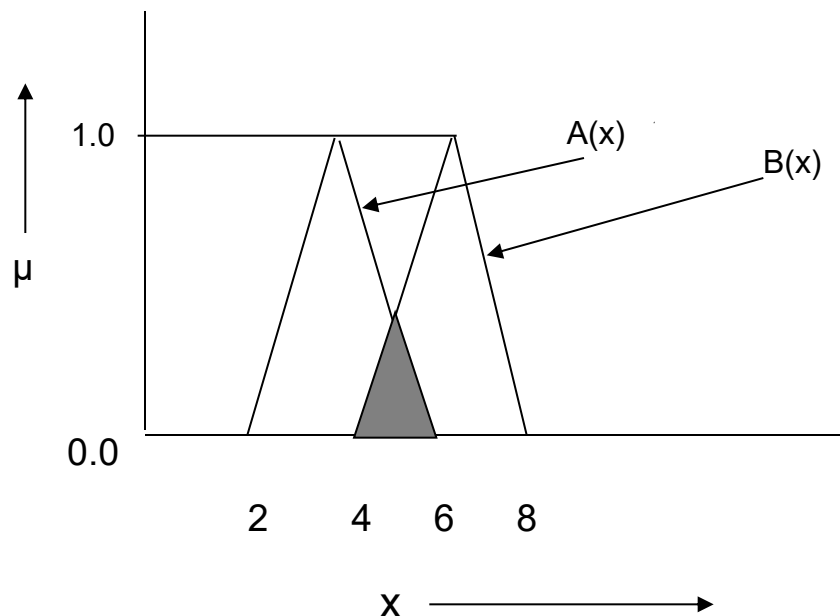
- **Example:**

Let us consider the following two fuzzy sets:

$$A(x) = \{(X_1, 0.1), (X_2, 0.2), (X_3, 0.3), (X_4, 0.4)\}$$

$$B(x) = \{(X_1, 0.5), (X_2, 0.7), (X_3, 0.8), (X_4, 0.9)\}$$

$$\mu_{(A \cap B)}(x_1) = \min\{\mu_A(x_1), \mu_B(x_1)\} = \min\{0.1, 0.5\} = 0.1$$





Union of fuzzy sets

- $\mu_{(A \cup B)}(x) = \max\{\mu_A(x), \mu_B(x)\}$

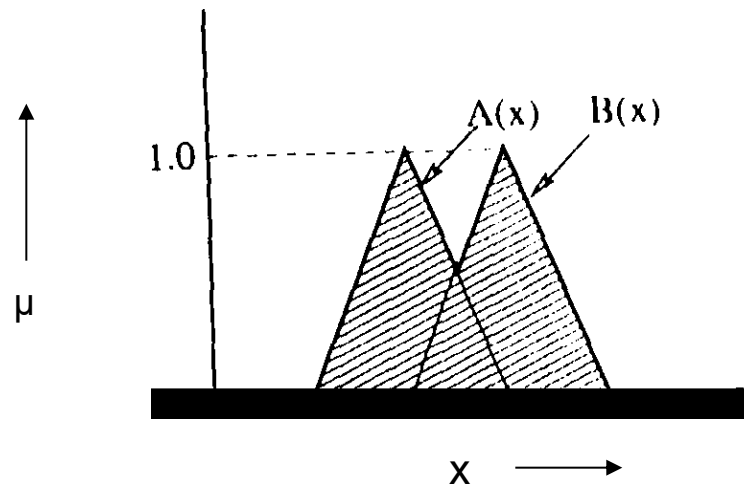
- **Example:**

Let us consider the following two fuzzy sets:

$$A(x) = \{(X_1, 0.1), (X_2, 0.2), (X_3, 0.3), (X_4, 0.4)\}$$

$$B(x) = \{(X_1, 0.5), (X_2, 0.7), (X_3, 0.8), (X_4, 0.9)\}$$

$$\mu_{(A \cup B)}(x_1) = \max\{\mu_A(x_1), \mu_B(x_1)\} = \max\{0.1, 0.5\} = 0.5$$





Algebraic product of fuzzy sets

- $A(x).B(x)=\{(x,\mu_A(x)*\mu_B(x)), x \in X\}$
- Example:

Let us consider the following two fuzzy sets:

$$A(x)= \{(X_1,0.1), (X_2,0.2), (X_3,0.3), (X_4,0.4)\}$$

$$B(x)= \{(X_1,0.5), (X_2,0.7), (X_3,0.8), (X_4,0.9)\}$$

$$\text{Now } A(x).B(x)=\{(X_1,0.05), (X_2,0.14), (X_3,0.24), (X_4,0.36)\}$$

Multiplication of a fuzzy set by a crisp number



- $d.A(x) = \{(x, d \times \mu_A(x)), x \in X\}$

- Example:

$$A(x) = \{(X_1, 0.1), (X_2, 0.2), (X_3, 0.3), (X_4, 0.4)\}$$

$$\text{if } d=0.2 \text{ then } d.A(x) = \{(X_1, 0.02), (X_2, 0.04), (X_3, 0.06), (X_4, 0.08)\}$$



Power of a fuzzy set

- The p-th power of a fuzzy set $A(x)$ yields another fuzzy $A^P(x)$, whose membership value can be determined as follows:
- $\mu_{A^p}(x) = \{\mu_A(x)\}^p, x \in X$
- If p is set equal to 2 and $\frac{1}{2}$ then $A^P(x)$ is called **concentration** and **dilation**, respectively.
- Example:

$$A(x) = \{(X_1, 0.1), (X_2, 0.2), (X_3, 0.3), (X_4, 0.4)\}$$

Now for $p=2$,

$$A^2(x) = \{(X_1, 0.01), (X_2, 0.04), (X_3, 0.09), (X_4, 0.16)\}$$



Algebraic sum of two fuzzy Sets

- $A(x)+B(x)=\{(x,\mu_{A+B}(x)), x \in X\}$

where $\mu_{A+B}(x)=\mu_A(x)+\mu_B(x)-\mu_A(x).\mu_B(x)$

- Example:

Let us consider the following two fuzzy sets:

$$A(x)= \{(X_1,0.1), (X_2,0.2), (X_3,0.3), (X_4,0.4)\}$$

$$B(x)= \{(X_1,0.5), (X_2,0.7), (X_3,0.8), (X_4,0.9)\}$$

$$\text{Now } A(x)+B(x)=\{(X_1,0.55), (X_2,0.76), (X_3,0.86), (X_4,0.94)\}$$



Bounded sum of two fuzzy sets

- $A(x) \oplus B(x) = \{(x, \mu_{A \oplus B}(x)), x \in X\}$

where $\mu_{A \oplus B}(x) = \min\{1, \mu_A(x) + \mu_B(x)\}$

- Example:

Let us consider the following two fuzzy sets:

$A(x) = \{(X_1, 0.1), (X_2, 0.2), (X_3, 0.3), (X_4, 0.4)\}$

$B(x) = \{(X_1, 0.5), (X_2, 0.7), (X_3, 0.8), (X_4, 0.9)\}$

Now $A(x) \oplus B(x) = \{(X_1, 0.6), (X_2, 0.9), (X_3, 1.0), (X_4, 1.0)\}$



Algebraic difference of two fuzzy Sets

- $A(x)-B(x)=\{(x,\mu_{A-B}(x)), x \in X\}$

where $\mu_{A-B}(x)=\mu_{A \cap \bar{B}}(x)$

- Example:

Let us consider the following two fuzzy sets:

$A(x)=\{(X_1,0.1), (X_2,0.2), (X_3,0.3), (X_4,0.4)\}$

$B(x)=\{(X_1,0.5), (X_2,0.7), (X_3,0.8), (X_4,0.9)\}$

Now $\bar{B}(x)=\{(X_1,0.5), (X_2,0.3), (X_3,0.2), (X_4,0.1)\}$

Therefore $A(x)-B(x)=\{(X_1,0.1), (X_2,0.2), (X_3,0.2), (X_4,0.1)\}$

Bounded difference of two fuzzy Sets



- $A(x) \ominus B(x) = \{(x, \mu_{A \ominus B}(x)), x \in X\}$

where $\mu_{A \ominus B}(x) = \max\{0, \mu_A(x) + \mu_B(x) - 1\}$

- Example:

Let us consider the following two fuzzy sets:

$$A(x) = \{(X_1, 0.1), (X_2, 0.2), (X_3, 0.3), (X_4, 0.4)\}$$

$$B(x) = \{(X_1, 0.5), (X_2, 0.7), (X_3, 0.8), (X_4, 0.9)\}$$

$$\text{Now } A(x) \ominus B(x) = \{(X_1, 0), (X_2, 0), (X_3, 0.1), (X_4, 0.3)\}$$



Cartesian product of two fuzzy sets

- Denoted by $A(x)XB(y)$ such that $x \in X$ and $y \in Y$
- $\mu_{AXB}(x,y) = \min(\mu_A(x), \mu_B(y))$
- Example:

Let us consider the following two fuzzy sets:

$$A(x) = \{(x_1, 0.2), (x_2, 0.3), (x_3, 0.5), (x_4, 0.6)\}$$

$$B(y) = \{(y_1, 0.8), (y_2, 0.6), (y_3, 0.3)\}$$

$$A(x)XB(y) = \begin{bmatrix} 0.2 & 0.2 & 0.2 \\ 0.3 & 0.3 & 0.3 \\ 0.5 & 0.5 & 0.3 \\ 0.6 & 0.6 & 0.3 \end{bmatrix}$$



Composition of fuzzy relations

- Let $A = [a_{ij}]$ and $B = [b_{jk}]$ be two fuzzy relations expressed in the matrix form.
- Composition $C = A \circ B$, or $[c_{ik}] = [a_{ij}] \circ [b_{jk}]$
where $c_{ik} = \max[\min(a_{ij}, b_{jk})]$
- Example:

$$A = [a_{ij}] = \begin{bmatrix} 0.2 & 0.3 \\ 0.5 & 0.7 \end{bmatrix}$$

$$B = [b_{jk}] = \begin{bmatrix} 0.3 & 0.6 & 0.7 \\ 0.1 & 0.8 & 0.6 \end{bmatrix}$$

$$C = \begin{bmatrix} 0.2 & 0.3 & 0.3 \\ 0.3 & 0.7 & 0.6 \end{bmatrix}$$

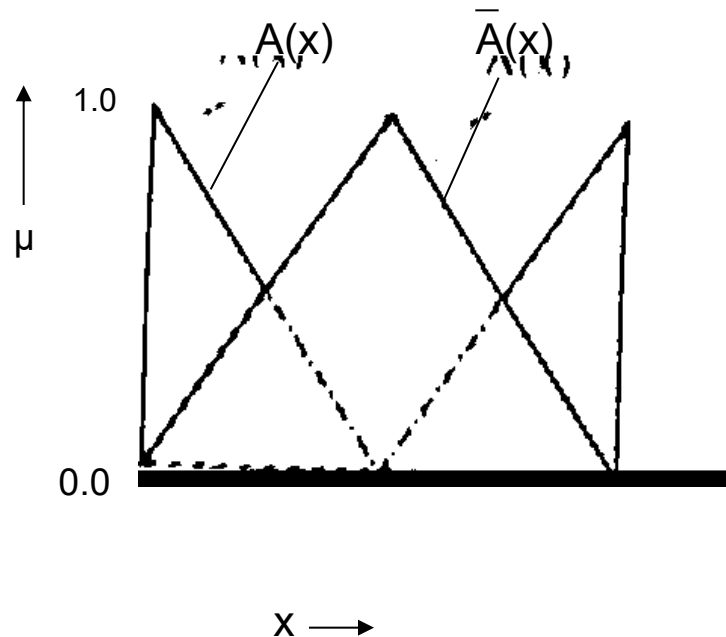


Properties of Fuzzy Sets



Law of excluded middle

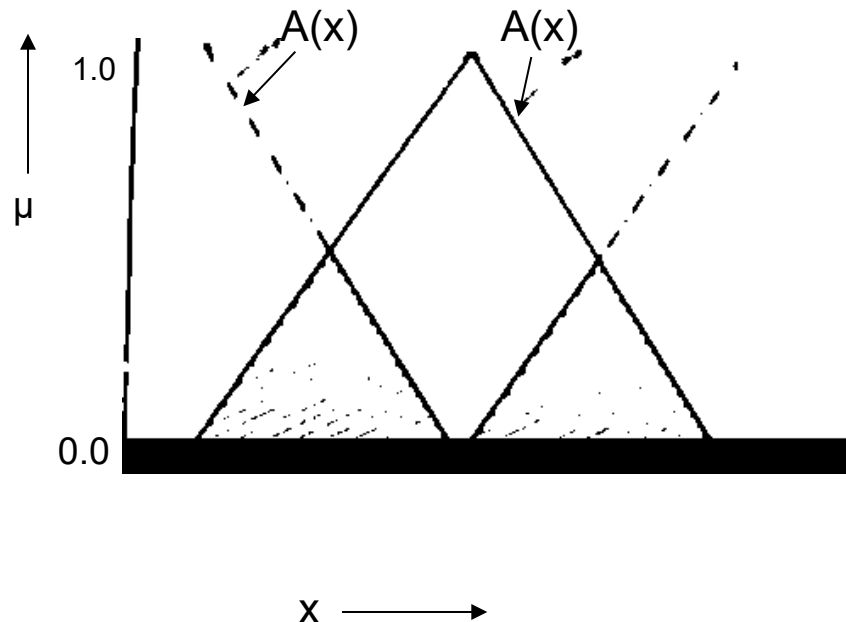
- In crisp set, $A \cup \bar{A} = X$.
- However, according to the fuzzy set, $A \cup \bar{A} \neq X$.





Law of contradiction

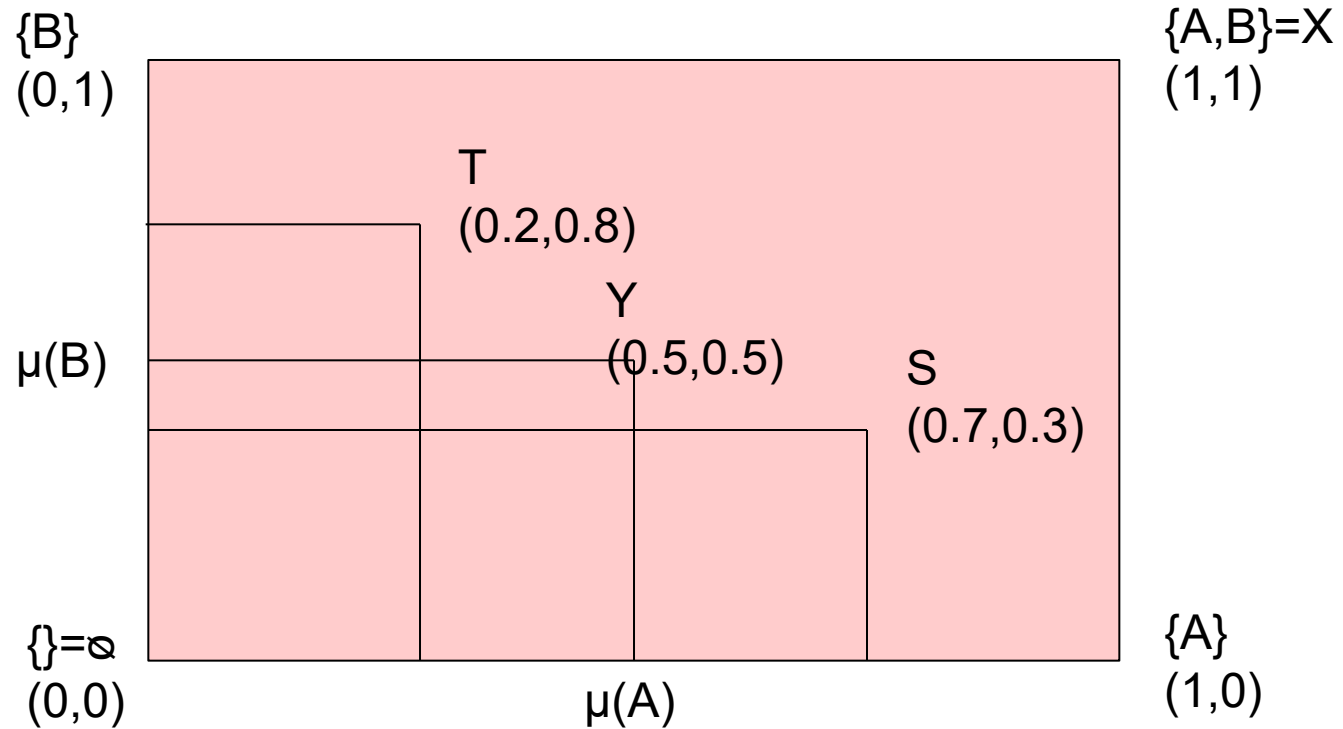
- In crisp set, $A \cap \bar{A} = \phi$.
- However, according to the fuzzy set, $A \cap \bar{A} \neq \phi$.





Geometry of fuzzy sets

- As shown in Figure, a fuzzy set can be given an interesting geometrical interpretation [3].





Geometry of fuzzy sets

- Consider a universe of two objects, A and B .
- In this geometry, we assume that each axis describes the degrees of membership of objects A and B respectively in a set P which can be either classical or fuzzy.
- In the geometrical representation of Figure, coordinate $(0,0)$ is interpreted as $A/0 + B/0$, implying that neither A nor B are members of P .
- The origin represents $P = \emptyset$ or the null set.
- Coordinate $(1,1)$ is interpreted as $A/1 + B/1$, where both A and B are members of P , and this point therefore represents the universal set $P = \{A, B\} = X$.
- Based on this reasoning, coordinates $(1,0)$ and $(0,1)$ are interpreted as representing the singleton sets $\{A\}$ and $\{B\}$ respectively.



Geometry of fuzzy sets

- The classical power set $2^X = \{\{\}, \{A\}, \{B\}, \{A,B\}\}$ is represented by the four corners of the unit square and is denoted by B^2 .
- The point S in Figure with coordinates $(0.7, 0.3)$ represents a fuzzy set $S = A/0.7 + B/0.3$ with the connotation: A and B are members of S to degrees 0.7 and 0.3 respectively.
- In the present example, the points that lie within the square denoted by $I^2 = [0, 1] \times [0, 1]$, represent different fuzzy sets.
- The set of all such possible fuzzy sets—the infinite points in I^2 —comprise the fuzzy power set F^X .
- Note that since $B^2 \subset I^2$, classical sets are special cases of fuzzy sets.



Fuzzy entropy theorem

- The degree of fuzziness of a fuzzy set can be measured by its *entropy*. In the geometric visualization, this corresponds inversely to the distance between the representation of the set and the center of the unit square. The set Y in Figure has the maximum possible entropy. The vertices represent the crisp sets and have the lowest entropy, that is, zero. Note that the fuzzy concept of entropy is mathematically different from the entropy concept in physics or information theory. Some authors prefer to use terms like *index of fuzziness* or also *crispness*, *certitude*, *ambiguity*, etc.



Fuzzy Measures

- Since the **evidence** of judgment is rarely perfect and measurement error is unavoidable in most physical sciences, some uncertainty usually prevails.
- To represent this kind of uncertainty, known as ambiguity , we assign a value in the unit interval $[0,1]$ to each possible crisp set to which the element in question might belong.
- This value represents the degree of evidence or belief or certainty of the element in the set.
- Such representation of uncertainty is known as **fuzzy measure**.



Evidence theory

- The information in many problems of design under uncertainty, especially those involving **reducible (epistemic) uncertainty**, is imprecise.
- Reducible uncertainty is uncertainty due to lack of knowledge, as opposed to **random (aleatory) uncertainty**, which is due to inherent variability in a physical phenomenon.
- It is called reducible, because it can be reduced or eliminated if one collects information.
- The uncertainty in the probability of getting heads in one flip of a bent coin is reducible, because it is due to lack of knowledge. It can be reduced if we conduct experiments.



Evidence theory

- On the other hand there is aleatory uncertainty when flipping a coin even if we know the probability of events; heads and tails. This type of uncertainty cannot be reduced even if we conduct n experiments, where n is a very large number. For example, we know the probability of heads and tails in a fair coin is 0.5. But each time you flip it, you are uncertain about the output. So this uncertainty is also called irreducible uncertainty.
- Theories for modelling epistemic uncertainty, include Coherent Upper and Lower Previsions, Possibility Theory, **Evidence Theory**, the Transferable Belief Model and Bayesian Theory.
- Information about epistemic uncertainty is usually in the form of **intervals**.
- For example, if we show a glass jar containing beans to a person and ask her how many beans are in the jar, then she is more likely to give a range rather than a precise number.



Evidence theory

- The performance of a system is characterized by variable Y , which is a function of uncertain variables $X_1, \dots, X_m$.
- We know the relation between the performance variable Y and variables $X_1, \dots, X_m$, $y = f(x_1, \dots, x_m)^2$
- We have information about the values of variables $X_1, \dots, X_m$, which is in the form of intervals obtained from n experts.
- These intervals have the form $[x^l_{ij}, x^u_{ij}]$ where subscript i specifies the variable, and j specifies the expert.
- This maximum probability will also be called Plausibility.
- This minimum probability will also be called Belief.



Fuzzy sets and Fuzzy Measures

- Consider a group of people X .
- For fuzzy sets, the age of the person $x \in X$ is known and consider x to be “Old” with a membership grade of $\mu_A(x)$, where A is a fuzzy set of “Old” people.
- For fuzzy Measures, the age of the person $x \in X$ is unknown, and by looking at the person, we consider him or her to be in the crisp set consisting of people who are more than 50 years old with a measure $g(A)$. The value $g(A)$ is a fuzzy measure that assigns a value to each possible crisp set (i.e., each age group of people) to which the person in question belongs.



Fuzzy Measures

- In general, a fuzzy measure is defined by a (set) function
 $g: \mathcal{P}(X) \rightarrow [0,1]$
- Which assigns to each crisp subset of a universe of discourse X a number in the unit interval $[0,1]$, where $\mathcal{P}(X)$ is the power set of X i.e. all possible subset of X . For example if $X = \{a, b\}$ then
 $\mathcal{P}(X) = \{a\}\{b\}\{a,b\}\{\emptyset\}$
- A fuzzy measure is a (set) function
 $g: \mathcal{F} \rightarrow [0,1]$

Where $\mathcal{F} \subset \mathcal{P}(X)$ is a family of crisp subsets of X and is a Borel field or a σ -field and g satisfies the following three axioms of fuzzy measures.



Fuzzy Measures

- **Axiom g1 (Boundary conditions):**

$$g(\emptyset) = 0 \text{ and } g(X) = 1$$

- **Axiom g2 (Monotonicity):**

For every crisp set $A, B \in \mathcal{P}(X)$, if $A \subseteq B$, then $g(A) \leq g(B)$

- **Axiom g3 (Continuity):**

For every sequence $(A_i \in \mathcal{P}(X))$ $i \in \mathbb{N}$ of subsets of X , if either $A_1 \subseteq A_2 \subseteq \dots$, then

$$\lim_{i \rightarrow \infty} g(A_i) = g(\lim_{i \rightarrow \infty} A_i)$$

- Where $\mathbb{N}$ is the set of all positive integers.

A **Borel field** $\mathcal{B}$ or a field satisfies these properties:

- $\emptyset \in \mathcal{B}$ and $X \in \mathcal{B}$

- If $A \in \mathcal{B}$, then $\bar{A} \in \mathcal{B}$

- $\mathcal{B}$ is closed under the operation of set union; that is, if $A \in \mathcal{B}$ and $B \in \mathcal{B}$, then $A \cup B \in \mathcal{B}$



Belief Measures

- The Belief Measures is a special type of fuzzy Measures that satisfies Axioms g1-g3 and an axiom of subadditivity (Axiom g4).

$$\text{Bel}: \mathcal{B} \rightarrow [0,1]$$

- Axiom g4:

$$\text{Bel}(A_1 \cup A_2 \cup \dots \cup A_n) \geq \sum_i \text{Bel}(A_i) - \sum_{i < j} \text{Bel}(A_i \cap A_j) + \dots + (-1)^{n-1}$$

$$\text{Bel}(A_1 \cap A_2 \cap \dots \cap A_n)$$

For every $n \in \mathcal{N}$ and every collection of subsets of X

- For each crisp set $A \in \mathcal{B} (\subset \mathcal{P}(X))$, $\text{Bel}(A)$ is interpreted as the degree of belief that a given element of X belongs to the set A . Axiom g4 indicates that knowledge of $\text{Bel}(A)$ and $\text{Bel}(B)$ is not always sufficient to calculate $\text{Bel}(A \cup B)$.



Belief Measures

- For $n=2$, Axiom g4 has the form

$$\text{Bel}(A_1 \cup A_2) \geq \text{Bel}(A_1) + \text{Bel}(A_2) - \text{Bel}(A_1 \cap A_2)$$

- When the sets $A_1, A_2, \dots, A_n$ in X are pairwise disjoint (i.e., $A_i \cap A_j = \emptyset$ for all $i, j \in \{1, 2, \dots, n\}, i \neq j$). Axiom g4 requires that

$$\text{Bel}(A_1 \cup A_2 \cup \dots \cup A_n) \geq \text{Bel}(A_1) + \text{Bel}(A_2) + \dots + \text{Bel}(A_n)$$

- For $n=2$ and $A_1 \equiv A$ and $A_2 \equiv \bar{A}$, then

$$\text{Bel}(A_1 \cup A_2) = \text{Bel}(A \cup \bar{A})$$

$$\text{Bel}(A \cup \bar{A}) \geq \text{Bel}(A) + \text{Bel}(\bar{A}) - \text{Bel}(A \cap \bar{A})$$

Since $A \cup \bar{A} = X$ and $A \cap \bar{A} = \emptyset$, we have

$$\text{Bel}(X) \geq \text{Bel}(A) + \text{Bel}(\bar{A}) \text{ or } 1 \geq \text{Bel}(A) + \text{Bel}(\bar{A})$$

This indicates that a lack of belief in $x \in A$ does not imply a strong belief in $x \in \bar{A}$



Plausibility Measure

- $PI(A) = 1 - Bel(\bar{A})$ and $Bel(A) = 1 - PI(\bar{A})$
- Belief measures and plausibility measures are mutually dual and that one can be derived from the other.
- A plausibility measure is a function

$PI: \mathcal{B} \rightarrow [0,1]$

That satisfies Axioms g1-g3 and an axiom of subadditivity (Axiom g5).

$$PI(A_1 \cap A_2 \cap \dots \cap A_n) \leq \sum_i PI(A_i) - \sum_{i < j} PI(A_i \cup A_j) + \dots + (-1)^{n-1}$$

$$PI(A_1 \cup A_2 \cup \dots \cup A_n)$$

For every $n \in \mathcal{N}$ and every collection of subsets of X

For $n=2$ and $A_1 \equiv A$ and $A_2 \equiv \bar{A}$, then

$$PI(A \cap \bar{A}) \leq PI(A) + PI(\bar{A}) - PI(A \cup \bar{A}) \text{ so } PI(A) + PI(\bar{A}) \geq 1$$



Basic Probability Assignment

- A Basic Probability Assignment m is a set function

$m: \mathcal{B} \rightarrow [0,1]$ such that $m(\emptyset) = 0$ and $\sum_{A \in \mathcal{B}} m(A) = 1$

Is interpreted as the degree of evidence indicating that a specific element of X belongs to the set A but not to any special subset of A . It is not a fuzzy measure.

- $\text{Bel}(A) = \sum_{B \subseteq A} m(B)$
- $\text{Pl}(A) = \sum_{B \cap A \neq \emptyset} m(B)$
- $\text{Pl}(A) \geq \text{Bel}(A) \geq m(A)$ for all $A \in \mathcal{B}$
- $m(A) = \sum_{B \subseteq A} (-1)^{|A-B|} \text{Bel}(B)$ for all $A \in \mathcal{B}$
- $m(A) = \sum_{B \subseteq A} (-1)^{|A-B|} [1 - \text{Pl}(\bar{B})]$ for all $A \in \mathcal{B}$



Basic Probability Assignment

- $m(A)$ measures the belief that the element in question ($x \in X$) belongs to the set A alone, not the total belief that the element commits to A .
- $Bel(A)$ indicates the total evidence that the element ($x \in X$) belongs to the set A and to the various special subsets of A .
- $Pl(A)$ represents the total evidence or belief that the element ($x \in X$) belongs to the set A or to any of the various special subsets of A plus the additional evidence or belief associated with sets that overlap with A .



Problem

Focal element	m (.)
P	0.05
B	0.05
E	0.05
PUB	0.15
PUE	0.1
BUE	0.05
PUBUE	0.55

Find out Bel and PI of focal elements.



Possibility & necessary measures

- Consonant belief and plausibility measures are usually referred to as necessity and possibility measures and denoted as N and π , respectively.
- $N: \mathcal{B} \rightarrow [0,1]$
- $\pi: \mathcal{B} \rightarrow [0,1]$
- Both N and π satisfy axioms g1-g3 of fuzzy measures and the following additional axiom (axiom g7)
- $N(A \cap B) = \min(N(A), N(B))$
- $\pi(A \cup B) = \max(\pi(A), \pi(B)), \forall A, B \in \mathcal{B}$
- Since necessity and possibility measures are special subclasses of belief and plausibility measures, respectively, they are related to each other by
- $\pi(A) = 1 - N(\bar{A})$
- $N(A) = 1 - \pi(\bar{A})$



References

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Thank you

Questions?